

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following questions.

(6x5=30)

1. What is Neumann's function? Give its graphical representation of Neumann's function at $v=0$, $v=1$, and $v=2$.

2. Find Green's Function for $\nabla^2 \psi = q$.

3. Find Fourier sine transform of e^{-ax} .

4. Find Fourier transform of slit function defined as

$$f(t) = 1 \quad \text{if } -a \leq t \leq a$$

$$f(t) = 0 \quad \text{otherwise}$$

5. Show that electrostatic potential produced by charge q at $z=a$ for $r < a$ is;

$$\varphi(r) = \frac{q}{4\pi\epsilon_0 a} \sum_{n=0}^{\infty} \left(\frac{r}{a}\right)^n P_n(\cos\theta)$$

6. Prove that polynomial $P_3(x) = \frac{1}{2}(5x^3 - 3x)$ is the solution of Legendre's differential equation.

Q.2. Solve the following questions.

(5x6=30)

1. Use the direct series substitution on the Legendre equation;

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

And verify that;

$$y_{even} = a_0 \left[1 - \frac{n(n+1)}{2!} x^2 + \frac{n(n+1)(n-2)(n+3)}{4!} x^4 + \dots \right]$$

Case 1: $y(1) = 0$
 $y(0) = 1$

2. Prove that Bessel functions show orthogonality property defined as

$$\int_0^1 J_n(ax) J_n(\beta x) dx = 0 \quad \text{if } a \neq \beta$$

$$= -a \leq t \leq a$$

3. Find the Fourier transform of Gaussian probability function $f(x) = N e^{-ax^2}$.

4. Find the Eigen functions and Eigen values of Sturm - Liouville's system $Y'' + \lambda Y$ with boundary conditions $Y(0) = 0$ and $Y'(\pi/2) = 0$.

5. Show that $\frac{d^2}{dx^2} G(x, x') + k^2 G = -\delta(x - x')$ gives us a Green's function

$$G(x, x') = \frac{\text{Sink}(1-x)\text{Sink}x}{k\text{Sink}}$$

where the boundary conditions are $G(0, x') = G(1, x') = 0$.

Past Paper ADP 2022

Q: 1
i) What is Neuman function.
Give graphical representation
of Neuman function
at $v=0$, $v=1$, and $v=2$.

As an alternate approach we
take particular linear combination
of $J_v(x)$ and $J_{-v}(x)$

$$N_v(x) = \frac{\cos v\pi J_v(x) - J_{-v}(x)}{\sin v\pi}$$

This is Neuman function.
For non integral v , $N_v(x)$ clearly
satisfies Bessel function.

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2) Find Green function
for $\nabla^2 \psi = q$

Let $G(\vec{r}, \vec{r}')$ be green
function for differential
operator ∇^2 , then

$$\nabla^2 G(\vec{r}, \vec{r}') = \begin{cases} \delta(\vec{r} - \vec{r}') & \text{if } \vec{r} = \vec{r}' \\ 0 & \text{if } \vec{r} \neq \vec{r}' \end{cases}$$

For $\vec{r} \neq \vec{r}'$

$$\nabla^2 \psi = 0$$

In spherical coordinates

$$\frac{1}{r^2} \frac{\partial^2}{\partial r^2} (r\psi) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} = 0$$

Due to symmetry, ψ depends
only on "r" so

$$\frac{1}{r^2} \frac{\partial^2}{\partial r^2} (r\psi) = 0$$

$$\Rightarrow \frac{\partial^2}{\partial r^2} (r\psi) = 0 \Rightarrow \frac{\partial}{\partial r} (r\psi) = A$$

$$\Rightarrow r\psi = Ar + B$$

$$\psi = A + \frac{B}{r}$$

Apply boundary condition

$\psi \rightarrow 0$ as $r \rightarrow 0$ so $A=0$

$$\boxed{A=0}$$

To find value of B , we integrate $\nabla^2 \psi = \delta(\vec{r} - \vec{r}')$ throughout volume sphere.

$$\int \nabla^2 \psi d\psi = \int \delta(\vec{r} - \vec{r}') d\psi$$

Apply divergence theorem on left side of above eq.

$$\int \nabla \psi \cdot d\vec{S} = 1 \Rightarrow \int \nabla \left(\frac{B}{r} \right) \cdot d\vec{S} = 1$$

$$= \int B \nabla \left(\frac{1}{r} \right) \cdot d\vec{S} = 1$$

$$= \int B \left(-\frac{\vec{r}}{r^2} \right) \cdot d\vec{S} = 1$$

$$= -B \int \frac{1}{r^2} \vec{r} \cdot d\vec{S} = 1 \Rightarrow -B \int \frac{\vec{r} \cdot d\vec{S}}{r^2} = 1$$

$$= -B \times 4\pi = 1$$

$$\boxed{B = -\frac{1}{4\pi}}$$

$$\text{So } \psi(r) = -\frac{1}{4\pi r}$$

when charge q is present Then

$$\psi(r) = -\frac{q}{4\pi r}$$

So required Green function is

$$G(\vec{r}, \vec{r}') = \frac{q}{4\pi (\vec{r} - \vec{r}')$$

3) Find Fourier sine transform of $f(x) = e^{-ax}$

As we know that Fourier sine transform of a function is

$$F_s(k) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin kx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} \sin kx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} e^{ikx} \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-(a-ik)x} \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left| \frac{e^{-(a-ik)x}}{-(a-ik)} \right|_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \cdot \frac{1}{-(a-ik)} \left[e^{-(a-ik)x} \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \cdot \frac{1}{-(a-ik)} (1)$$

$$= \sqrt{\frac{2}{\pi}} \cdot \frac{1}{a-ik}$$

4) Find Fourier transform of Slit function:

$$\begin{cases} f(t) = 1 & ; -a \leq t \leq a \\ f(t) = 0 & ; \text{otherwise} \end{cases}$$

As Fourier transform of a function is defined as

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{ikx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{-a} f(x) e^{ikx} dx + \frac{1}{\sqrt{2\pi}} \int_{-a}^a f(x) e^{ikx} dx + \frac{1}{\sqrt{2\pi}} \int_a^{\infty} f(x) e^{ikx} dx$$

$$F(k) = 0 + \frac{1}{\sqrt{2\pi}} \int_{-a}^a f(x) e^{ikx} dx + 0$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{ikx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{ikx}}{ik} \right]_{-a}^a$$

$$= \frac{1}{\sqrt{2\pi}} ik \left[\frac{e^{ika}}{ik} - \frac{e^{-ika}}{ik} \right] = \frac{2}{\sqrt{2\pi}} \left[\frac{e^{ika} - e^{-ika}}{2i} \right]$$

$$= \frac{2}{\sqrt{2\pi}} \cdot \frac{\sin ak}{k}$$

$$= \sqrt{\frac{2}{\pi}} k \sin ak \quad \Rightarrow \quad \sqrt{\frac{2}{\pi}} \frac{\sin ak}{k}$$

5) show that
Electrostatic Potential
by charge q at $z=a$
for $z < a$ is

$$\phi(r) = \frac{q}{4\pi\epsilon_0 a} \sum_{n=0}^{\infty} \left(\frac{r}{a}\right)^n P_n(\cos\theta)$$

Consider a point charge q at A
on z -axis. Suppose distance of charge
 q from origin " a " we want to
calculate electric Potential at a
Point $P(x, y, z)$ at distance r .
Electric Potential at Point P
due to charge q is given by

$$V = \frac{q}{4\pi\epsilon_0} \cdot \frac{1}{r_1} \quad \text{--- (1)}$$

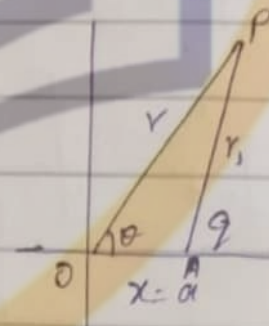
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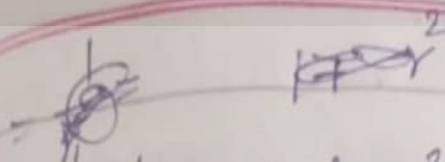
by law of cosines

$$r_1^2 = r^2 + a^2 - 2ar\cos\theta$$

$$r_1 = (r^2 + a^2 - 2ar\cos\theta)^{\frac{1}{2}}$$

$$\frac{1}{r_1} = (r^2 + a^2 - 2ar\cos\theta)^{-\frac{1}{2}}$$





$$\frac{1}{r_1} = (r^2 + a^2 - 2ar \cos \theta)^{-\frac{1}{2}}$$

$$\frac{1}{r_1} = \frac{1}{a} \left(\frac{r^2}{a^2} + 1 - \frac{2r \cos \theta}{a} \right)^{-\frac{1}{2}}$$

$$\frac{1}{r_1} = \frac{1}{a} \left(1 + \frac{r^2}{a^2} - \frac{2r \cos \theta}{a} \right)^{-\frac{1}{2}}$$

$$\frac{1}{r_1} = \frac{1}{a} \sqrt{1 + \frac{r^2}{a^2} - \frac{2r \cos \theta}{a}}$$

Comparing with generating function of Legendre polynomial

$$\sum_{l=0}^{\infty} P_l(x) t^l = \frac{1}{\sqrt{1-2xt+t^2}}$$

we find

$$\frac{1}{r_1} = \frac{1}{a} \sum_{l=0}^{\infty} \left(\frac{r}{a} \right)^l P_l \cos \theta$$

Put in (1)

$$V(r) = \frac{q}{4\pi\epsilon_0 a} \sum_{l=0}^{\infty} \left(\frac{r}{a} \right)^l P_l \cos \theta$$

Proved

2) Prove that Bessel function show orthogonally property defined as:

$$\int_0^a x J_n(ax) J_n(Bx) dx = 0$$

if $a \neq B$

$$= -a < + < a$$

when $a \neq B$

As Bessel diff. eq. is

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0$$

$$= x^2 y'' + x y' + (x^2 - n^2) y = 0$$

As $J_n(ax)$ and $J_n(Bx)$ are Bessel fn.

So they satisfy Bessel eq.

$$x^2 J_n''(ax) + x J_n'(ax) + (a^2 x^2 - n^2) J_n(ax) = 0 \quad \text{--- (1)}$$

and $x^2 J_n''(Bx) + x J_n'(Bx) + (B^2 x^2 - n^2) J_n(Bx) = 0 \quad \text{--- (2)}$

Multiply eq. (1) by $J_n(Bx)$ and (2) by $J_n(ax)$

and subtracting we have

$$x^2 [J_n(Bx) J_n''(ax) - J_n(ax) J_n''(Bx)] + x [J_n(Bx) J_n'(ax) - J_n(ax) J_n'(Bx)] + x^2 (a^2 - B^2) J_n(ax) J_n(Bx) = 0$$

Divide by x

$$x [J_n(Bx) J_n''(ax) - J_n(ax) J_n''(Bx)] + [J_n(Bx) J_n'(ax) - J_n(ax) J_n'(Bx)] + x (a^2 - B^2) J_n(ax) J_n(Bx) = 0$$

$$\Rightarrow \frac{d}{dx} \left[x \{ J_n(\alpha x) J_n'(\beta x) - J_n(\beta x) J_n'(\alpha x) \} \right] + x(\alpha^2 - \beta^2) J_n(\alpha x) J_n(\beta x)$$

$$= \frac{d}{dx} \left[x \{ J_n(\beta x) J_n'(\alpha x) - J_n(\alpha x) J_n'(\beta x) \} \right] = (\beta^2 - \alpha^2) x J_n(\alpha x) J_n(\beta x)$$

Integrating w.r.t "x" we get b/w limits 0 and a we get

$$\int_0^a (\beta^2 - \alpha^2) x J_n(\alpha x) J_n(\beta x) dx = \left[x \{ J_n(\beta x) J_n'(\alpha x) - J_n(\alpha x) J_n'(\beta x) \} \right]_0^a$$

$$\beta^2 - \alpha^2 \int_0^a x J_n(\alpha x) J_n(\beta x) dx = a \{ J_n(\beta a) J_n'(\alpha a) - J_n(\alpha a) J_n'(\beta a) \}$$

Since α and β are roots of $J_n(ax) = 0$ i.e. $J_n(\alpha a) = 0$ and $J_n(\beta a) = 0$

So $(\beta^2 - \alpha^2) \int_0^a x J_n(\alpha x) J_n(\beta x) dx = 0$

3) Fourier Transform:-

of Gaussian Probability function $f(x) = N e^{-ax^2}$

By definition of Fourier Transform

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{ikx} dx$$

$$F(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} N e^{-ax^2} e^{ikx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} N e^{-ax^2 + ikx} dx \quad \text{--- (1)}$$

Now

$$\begin{aligned}
 -ax^2 + ikx &= -[ax^2 - ikx] \\
 &= -\left[ax^2 - ikx + \frac{k^2}{4a} - \frac{k^2}{4a}\right] \\
 &= -\left[\sqrt{a}x^2 - 2(\sqrt{a}x)\left(\frac{ik}{2\sqrt{a}}\right) + \left(\frac{ik}{2\sqrt{a}}\right)^2\right] - \frac{k^2}{4a}
 \end{aligned}$$

$$\therefore \left(\sqrt{a}x - \frac{ik}{2\sqrt{a}}\right)^2 - \frac{k^2}{4a} \Rightarrow -u^2 - \frac{k^2}{4a}$$

where

$$u = \sqrt{a}x - \frac{ik}{2\sqrt{a}}$$

$$du = \sqrt{a} dx$$

$$dx = \frac{du}{\sqrt{a}}$$

So eq. (1) becomes

$$\Psi(k) = \frac{N}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2 - \frac{k^2}{4a}} \frac{du}{\sqrt{a}}$$

$$= \frac{N}{\sqrt{2\pi}} \frac{e^{-\frac{k^2}{4a}}}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-u^2} du$$

$$= \frac{N}{\sqrt{2\pi}} \frac{e^{-\frac{k^2}{4a}}}{\sqrt{a}} \cdot \sqrt{\pi}$$

$$= \frac{N}{\sqrt{2a}} e^{-\frac{k^2}{4a}}$$

Q.E.D.

4) Find eigen function and eigen value of Sturm Liouville's system

$$y'' + \lambda y$$

boundary condition $y(0) = 0$
 $y'(\pi/2) = 0$

Sol: $y'' + \lambda y = 0$

Case I:- when $\lambda = 0$

Then $y'' = 0$, $y' = A$

$$y = Ax + B$$

Apply first boundary condition

$$y(0) = 0$$

$$\Rightarrow 0 = A(0) + B$$

$$\boxed{B = 0}$$

Apply second boundary condition

$$y'(\pi/2) = 0$$

$$y'(\pi/2) = A$$

$$\boxed{A = 0}$$

So for $\lambda = 0$, $y(x) = 0$

hence solution does not exist

Case II: λ is -ve

Put $\lambda = -n^2$

Then

$$y'' - n^2 y = 0$$

$$D^2 - n^2 = 0$$

n is non zero real number

$$D^2 \cdot y^2 = 0$$

$$D^2 = y^2$$

$$D = \pm n$$

$$\text{So } y(x) = A e^{nx} + B e^{-nx}$$

Apply first condition
 $y(0) = 0$

$$0 = A e^0 + B e^0$$

$$A + B = 0$$

$$A = -B, \quad |B = -A|$$

$$\text{So } y(x) = A e^{nx} - A e^{-nx}$$

$$y(x) = A (e^{nx} - e^{-nx})$$

Now $y'(x) = nA (e^{nx} + e^{-nx})$

Apply second boundary condition

$$y'\left(\frac{\pi}{2}\right) = nA (e^{n\pi/2} + e^{-n\pi/2})$$

$$0 = nA (e^{n\pi/2} + e^{-n\pi/2})$$

So $A = 0$ Then for

$$\lambda = -n^2 \quad y(x) = 0$$

Solution does not exist.

Case 3), λ is +ve

$$\text{put } \lambda = n^2$$

$$y'' + n^2 y = 0$$

$$D^2 + n^2 = 0$$

$$D = \pm in$$

$$y(x) = A \cos nx + B \sin nx$$

APP/1 first boundary cond:

$$y(x) = 0$$

$$y(0) = A \cos(n \cdot 0) + B \sin 0$$

$$\boxed{A = 0}$$

So $y(x) = B \sin nx$

Now

$$y'(x) = B \cos nx$$

APP/1 Second boundary condition

$$y'\left(\frac{\pi}{2}\right) = B \cos n \frac{\pi}{2}$$

For non trivial solution

Then $B \neq 0$

$$\cos n \frac{\pi}{2} = 0$$

$$\frac{n\pi}{2} = (2m-1) \frac{\pi}{2}$$

$$\boxed{n^2 = (2m-1)^2}$$

So for $\lambda = (2m-1)^2$

The solution is

$$y(x) = B \sin nx$$

Q

5)

Show that

$$\frac{d^2}{dx^2} G(x, x') + K^2 G = -\delta(x - x')$$

gives green function

$$G(x, x') = \frac{\sin K(1-x) \sin Kx}{K \sin K}$$

where boundary conditions

$$\text{are } G(0, x') = G(1, x') = 0$$

eq. can be written as

$$\left(\frac{d^2}{dx^2} + K^2 \right) G(x, x') = \delta(x - x')$$

$$\mathcal{L} G(x, x') = \delta(x - x')$$

$$\therefore d = \frac{d^2}{dx^2} + K^2$$

$$\Rightarrow \left(\frac{d^2}{dx^2} + K^2 \right) G(x, x') = 0$$

Characteristic eq.

$$-D^2 + K^2 = 0$$

$$D^2 = -K^2$$

$$D = \pm iK$$

So solution is

$$G(x, x') = A \cos Kx + B \sin Kx$$

Two cases arises

Case I) when $x < x'$

$$G_L(x, x') = A_1 \cos kx + B_1 \sin kx$$

Applying boundary condition:

$$G_L(0, x') = 0$$

$$0 = A \cos(0) + B \sin(0)$$

$$\boxed{A=0}$$

$$G_L(x, x') = B_1 \sin kx$$

$$G_L(x, x') = C \sin kx$$

Case II) when $x > x'$

$$G_R(x, x') = A_2 \cos kx + B_2 \sin kx$$

$$A_2 \cos kx + B_2 \sin kx = C$$

$$-A_2 \cos k = B_2 \sin k$$

$$B_2 = -A_2 \frac{\cos k}{\sin k}$$

$$G_R(x, x') = A_2 \cos kx - \frac{A_2 \cos k}{\sin k} \sin kx$$

$$= \frac{A_2 \sin k \cos kx - A_2 \cos k \sin kx}{\sin k}$$

$$= \frac{A_2}{\sin k} (\sin k \cos kx - \cos k \sin kx)$$

$$= \frac{A_2}{\sin k} \sin(k - kx)$$

$$D = \frac{\sin k(1-x')}{k \sin k} \cdot \frac{\sin kx'}{\sin k(1-x')}$$

$$D = \frac{\sin kx'}{k \sin k}$$

Put C and D in

① and ②

$$G_1(x, x') = \frac{\sin k(1-x') \sin kx}{k \sin k}$$

$$G_2(x, x') = \frac{\sin k(1-x) \sin kx'}{k \sin k}$$

Or

$$G(x, x') = \begin{cases} \frac{\sin k(1-x') \sin kx}{k \sin k} & \text{for } x < x' \\ \frac{\sin k(1-x) \sin kx'}{k \sin k} & \text{for } x > x' \end{cases}$$